

Project Management Analysis: Organising the ANZ Analytics Innovation Conference (2026)

A PERT and Cost-Time Trade-off Evaluation

Tianxiang Yang (618218476),
Langlang Shao (170528440),
Freya Ping (708571564),
Junwoo Lim (736846465)

Dr. Maya Thompson

OPSMGT 357: Project Management

20.05.2026

Contributor: Tianxiang Yang

Section 2: Executive Summary

This report evaluates the feasibility of delivering the ANZ Analytics Innovation Conference (2026) within the University Executive Committee's mandated 45-day deadline, applying the Program Evaluation and Review Technique (PERT) together with time–cost trade-off (crashing) analysis. The conference comprises fifteen interdependent activities across a precedence network containing five distinct paths, with activity durations expressed as three-point estimates to reflect planning uncertainty.

Under the baseline plan, the project is expected to take 46 days, exceeding the 45-day target. The probability of completing the project on time is only approximately 37.5%, indicating that the original plan is more likely to overrun the deadline than to meet it. The baseline total cost is approximately \$282,000.

To address this, five scenarios were analysed. Scenario 1 established that raising the probability of on-time completion to 95% requires compressing the schedule to 39 days, at an additional crash cost of approximately \$53,000 and a total cost of approximately \$305,600. Scenario 2 identified the cost-minimising schedule at 44 days and approximately \$279,100, but this option carries only an estimated 62% probability of meeting the deadline, leaving substantial residual risk. Scenario 3 determined the shortest achievable duration to be 36 days, though at a disproportionately high marginal cost. Most significantly, Scenario 4 demonstrated that expediting a single high-duration activity — venue booking and contracting (Task 4) — raises the on-time probability to approximately 50% at no additional cost, and reduces the cost of subsequently reaching 95% confidence to approximately \$33,000, a saving of around \$20,000 relative to crashing alone. Scenario 5 showed that additional crashing capacity for keynote speaker confirmation (Task 8) offers value only at the extreme of schedule compression and is therefore best retained as a contingency reserve.

Drawing on these findings, this report recommends that the Business School first pursue expedited approval for venue booking, as this structural change delivers the greatest improvement in schedule confidence at minimal cost. The School should target a 95% confidence schedule rather than the cost-minimising option, whose residual risk is not justified for an initiative of this strategic importance, and should avoid the minimum-duration schedule, whose marginal cost is excessive. Throughout execution, particular attention should be paid to the promotional campaign (Task 10), which is the largest single source of duration uncertainty, and to the near-critical path, which becomes binding as the schedule is compressed.

Together, these measures enable the Business School to deliver the conference with high confidence of meeting the 45-day deadline, at a total cost materially below that of the original 95% plan, while preserving the flexibility to respond to emerging risks.

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Section 4:Introduction

4.1 Project Context and Background

In our modern society, making strategic decisions relies on data-driven insights, artificial intelligence, and data analysis. To respond to this shift, the UoA Business School has established ANZ Analytics Innovation Conference (2026), a premier hybrid event that aims to bring together participants from industry leaders as well as academic researchers, public-sector practitioners and postgraduate students across Australia and New Zealand. The conference aims to solidify the University's position as a regional center of applied analytics knowledge and provide practical engagement opportunities with industry while integrating real-world project experience for students. For the first time, a project team of students studying OPSMGT 357 — Project Management has been assigned by the Business School to plan and coordinate the conference, with academic guidance provided by Dr. Maya Thompson¹ Director of Industry Engagement. The decision aligns with the School's nature of applied learning that is included in its Master of Business Analytics programme. This project involves fifteen interdependent activities across defining the theme, engaging the speakers, developing sponsors, managing the venue and hybrid logistics, technology setup for streamlining in-person and online interactions as well as marketing in advance and post-event evaluations — managed across multiple University and external stakeholders.

4.2 The Strategic Constraint

The conference is subject to a critical and non-negotiable constraint: the entire initiative — from initial planning through to event delivery — must be completed within 45 days. This deadline is dictated by the University Executive Committee, as the conference must occur prior to the start of the corporate recruitment season, when industry engagement reaches its annual peak. Compressing what would conventionally be a multi-month planning effort into 45 days creates considerable schedule pressure across multiple interdependent activities. The duration of each activity is provided as a three-point estimate (optimistic, most likely, and pessimistic), reflecting the inherent uncertainty in early-stage project planning. With fifteen activities forming a precedence network containing five distinct paths from project initiation to closure, the project exhibits the characteristic complexity of large-scale undertakings for which structured project management techniques have been developed (Project Management Institute [PMI], 2021)

4.3 Research Problem

Due to time constraints and uncertainty regarding the project duration, the University Executive Committee needs to conduct a project management assessment. The following questions must be addressed: What is the probability that the project will be completed within 45 days? Deciding the priority and financial cost of actions to speed up (if they are not

sufficient). Does the cost of additional schedule compression begin to exceed the savings in indirect expense provide a financial incentive to use? What does the project look like under different assumptions? If we had booked a venue, that has an additional lead-time? What if we have more flexibility to negotiate with keynote speakers etc. Some of these questions are not solvable through either experience or intuition alone, but require analytical techniques which explicitly account for task-level uncertainty and the time-cost tradeoff (e.g. Malcolm et al., 1959; Kelley, 1961).

4.4 Methodology and Scope

This report uses a combination of Program Evaluation and Review Technique (PERT) with time–cost trade-off analysis to examine the ANZ Analytics Innovation Conference plan and propose an optimum schedule. This aids in converting three-point duration estimates into expected times and variances, helps identify a critical path (the series of dependent activities that control the overall project duration), and allows computation of completion probabilities for given delivery or cost targets (Malcolm et al., 1959). It is then the basis for time–cost trade-off analysis, or crashing, that shows if and how duration can be shortened through additional resource allocations by considering the marginal cost of crashing versus indirect cost saved per day of reduced duration (Kelley, 1961). In the analysis, there are five scenarios given in the case. In scenario 1 for this use case, it investigates the crashing requirements to complete 95% probability within 45 days. Scenario 2 defines the cost–time optimum, namely the point in which total project cost incurs its minimum. Scenario 3 establishes the minimal duration and cost which can not be appreciated on a technical level. Scenarios 4 and 5 perform a type of sensitivity analyses on two planned modifications: speeding up the venue reservation process (Task 4) and increased crashing capacity for final confirmation of keynote speakers (Task 8). All calculations will be performed within the Excel model,

4.5 Report Structure

This report is structured as follows. The Literature Review lays the theoretical background to the analysis by studying the historical evolution of PERT and Critical Path Method (CPM), a review of mathematical theory behind time–cost trade-off, critical responses to assumptions underlying PERT, and pattern-matching applied examples from relevant fields. The Essay Body contains the full scenario analysis, starting from the baseline probability assessment and walking through each of the five scenarios sequentially (including migration of critical paths during crashing and higher-valued activities spanning multiple paths). The Strategic Recommendations Section synthesises the results to provide an integrated recommendation for the

Business School on the efficiency of achieving 45 days while outlining schedule weaknesses and recommending priorities to optimize resource deployment and minimize risk. Appendices include supporting tables, calculations, and visualisations.

5. Literature Review

5.1 Overview

This review lays the theoretical groundwork for the analytical methodology that this report follows. It is structured to follow the emergence of the pertinent methods both in a linear and thematic manner, starting with basis of project management as a field, followed by an exploration of origins in PERT and CPM, time–cost trade-off mathematics, and key aspects that articulate objections to some assumptions behind PERT. This review ends by placing these techniques in the context of modern project management standards, then identifies gaps that this report aims to fill. Its purpose is to explain the role of each technique but also, the assumptions and limitations that are important for understanding the scenario analysis further down.

5.2 Foundations of Project Management

Over the years project management has evolved into a relatively formal discipline with a well-defined body of knowledge and an extensive literature for practitioners. A project can be generally defined as a temporary initiative undertaken to create a unique output in constrained time, scope and resources (Project Management Institute [PMI], 2021).

According to Larson and Gray (2014), project management is defined as the managerial process that brings together planning, scheduling, and control activities spanning the life of a project — delivery comes from coordination and decision-making as much as technical analysis. A foundational planning tool within this process is the Work Breakdown Structure (WBS), which decomposes a project into progressively smaller and more manageable components, providing the basis for subsequent scheduling and resource allocation (Larson & Gray, 2014).

Beyond individual tools, the literature also addresses the organisational maturity required to apply project management effectively. Kerzner (2006) argues that sustained project success depends on embedding repeatable practices and structured methodologies across an organisation, rather than relying on the ad hoc efforts of individual teams. While Kerzner's focus is primarily organisational, the underlying principle — that disciplined, repeatable process improves project outcomes — is equally applicable to a small team applying these techniques to a defined task. These foundational perspectives frame the more specialised scheduling techniques discussed below.

5.3 The Origins of PERT

PERT emerged from a specific institutional context: the United States Navy's Polaris Fleet Ballistic Missile programme of the late 1950s. Malcolm, Roseboom, Clark, and Fazar (1959) introduced the technique as a means of evaluating and controlling progress in large-scale research and development programmes characterised by substantial uncertainty in activity

durations. A defining feature of their approach was the use of three-point time estimates — an optimistic time, a most likely time, and a pessimistic time — to capture the range of plausible outcomes for each activity, rather than relying on a single deterministic estimate (Malcolm et al., 1959).

From these three estimates, PERT derives an expected activity duration and a measure of variance, allowing the overall project completion time to be treated as a probabilistic quantity. The expected duration is conventionally calculated as a weighted average that gives greatest weight to the most likely estimate, while the variance is derived from the spread between the optimistic and pessimistic estimates. By aggregating activity variances along the critical path and invoking the Central Limit Theorem, the technique enables the probability of completing a project within a target deadline to be estimated. This probabilistic orientation distinguishes PERT from purely deterministic scheduling methods and makes it particularly suited to contexts, such as the present case, in which activity durations cannot be known with certainty. The technique's capacity to translate uncertainty into a quantifiable completion probability is directly relevant to the baseline assessment presented in the following section.

5.4 The Critical Path Method

While PERT was being developed, the Critical Path Method was also being implemented in a different environment for industrial project planning that had strong ties to work undertaken with DuPont corporation. An important early contribution was by Kelley (1961), who established the mathematical foundation for critical-path planning and scheduling, formalising how to find the sequence of activities that governs the overall minimum project duration. The most important of these paths is the critical path, which can be characterized as a longest duration path through the project network; activities on this path have no slack/their delay delays all activities (Kelley, 1961).

One of the central ideas that emerged from this analysis is that of float, or slack —the amount by which a noncritical activity can be delayed without delaying the overall completion date. While PERT was originally conceived as a way to manage uncertainty in an activity's duration, CPM is based around deterministic durations and the cost-duration relationship (Larson & Gray, 2014). There is a variety of PERT and CPM techniques used today, but they are often combined with techniques for deterministic estimation from PERT applied in the context of a more general probabilistic formulation embedded within the network structure of CCM. The present report provides insight into this integration by implementing three-point estimation in order to identify and analyse the critical path of the project network.

5.5 Time–Cost Trade-off and Crashing

Another input of Kelley (1961) was the formalization of a relationship between cost and duration to establish what is, in fact, referred to now as crashing. Crashing is the riskless shortening of a schedule by assigning more resources to each activity selected for time reduction at some extra direct cost (Larson & Gray, 2014). We segregate costs into direct,

which tend to increase when activities are compressed and indirect (or overhead), which tend to decrease as over project durations are shortened.

The purpose of time–cost trade-off analysis is to find a schedule with the least total cost achieved by balancing the marginal cost incurred by crashing an activity and the indirect cost saved per unit time. According to Larson and Gray (2014), the decision rule is to first crash activities on the critical path that have the lowest cost per unit of time saved, until marginal cost of further compression exceeds indirect savings or no more compression is technically possible. One aspect of this process is that compressing one path may actually force a non-critical path to become critical (measured as in terms total duration) — within an analysis perspective, this becomes difficult since you continually have to re-identify what the critical path is so crashing may continue into other paths. This is directly analysed in the subsequent scenario analysis covering several critical path migrations.

5.6 Critical Perspectives on PERT's Assumptions

Despite being used in documents all over the world, PERT has attracted continuous academic critique with a significant part of it focusing on inherent assumptions related to its probabilistic calculations. An early analytical study of PERT assumptions and sources of error in the technique, especially systematic approximations made to derive expected activity times and variances, was performed by MacCrimmon and Ryavec (1964). The most notable issue they have raised relates to their treatment of parallel paths: because PERT usually calculates the probability of completing on time relative only to 1 critical path, it can systematically underestimate expected project duration with multiple near–critical paths (all of which are equally likely a binding constraint in practice).

Sasieni (1986) also explored the derivation of the traditional PERT formula for expected activity time and highlighted its limiting assumptions. These criticisms do not discredit PERT, however they suggest little confidence in assuming the interpretations of its outputs, especially when there are many paths of comparable length; which holds true for this report's project network.

Critical Chain Project Management developed by Goldratt (1997) shifted the focus of project management away from the individual estimates for activity durations towards explicit scheduling buffers to safeguard the end-to-end schedule against uncertainty, partly in response to limitations of traditional network-based scheduling. A complete treatment of Critical Chain goes well beyond the scope of this review, but its principle of buffer-management somewhat parallels the contingent portions of a schedule set out in the strategic recommendations. Taken Collectively, these key perspectives highlight the value of viewing PERT outputs as informative estimates rather than exact predictions.

5.7 Contemporary Standards and Applications

The PMBOK Guide is the most recognized codification of modern project management practice in established standards. The seventh edition (PMI 2021) represents a significant turn in orientation from process-based to principles-based framework that could accommodate a

variety of delivery approaches. This evolution reflects the maturing of the discipline, adapting to an increasingly diverse project context.

Recent scholarship has explored the interrelations of such standards. Aydın (2023) provides a comparative analysis of the PMBOK guide seventh edition and ISO 21500:2021 by reviewing their definitions, structures and limitations to highlight areas of convergence and divergence between the two standards. This type of comparative work shows the ongoing evolution in the field and that there is no one framework adopted universally. For the purpose of this report, these standards act as a high-level professional context in which the more detailed methods of PERT and crashing are located.

5.8 Synthesis and Research Gap

As the literature reviewed above demonstrates, a foundational body of knowledge is maturing in its various applications. Malcolm et al.'s seminal contributions Techniques for probabilistic estimation and time–cost trade-off granted PERT its place as a project scheduling staple (1959, Kelley 1961), but subsequent criticisms raised concerns about the limitations of plotting estimations from assumptions underlying PERT. While contemporary standards delineate professional expectations in broad brush strokes, they provide little guidance on the integrated application of these techniques to practice-based projects with defined time constraints. There are relatively few published sources that effectively present an integrated demonstration of PERT, critical path analysis (CPA), crashing and sensitivity analysis as a consistent application to a single unified problem within the context of an explicit deadline shortfall. This is the research that is explored by using the ANZ Analytics Innovation Conference.

Section 6:Essay Body — 5 个 Scenario 分析(3,000)

Baseline Probability Assessment

Before any scenario can be tested, we need to know what the project looks like as planned. The PERT formula gives an expected time for each of the fifteen activities, which once summed across the network produces five paths from kickoff to closure. Their lengths are 42, 37, 45, 39, and 46 days. The longest, running through tasks 1, 4, 11, 12, 13, 14, and 15, is the critical path at 46 days, already one day over the 45-day mandate. So the plan is in trouble before uncertainty has even been factored in.

Adding the variances along this critical path gives 9.81, which means a standard deviation of 3.13 days. The Z-value is then -0.32, and the probability of finishing within 45 days works out to roughly 37.5 percent. The project would meet its deadline only one time in three. For a conference of this profile, that is not a number anyone can defend.

There is also a second issue that ends up shaping everything that follows. Path C, running through tasks 1, 3, 8, 9, 10, 14, and 15, comes to 45 days, only one day shorter than the critical path. The network has two long paths sitting almost on top of each other. As soon as the critical path is touched, the second one will catch up, so any crashing strategy effectively has to manage both at once.

Network Paths & Critical Path Identification																					
All 5 paths from start (Task 1) to finish (Task 15). Path length = sum of crashed durations along that path. Critical path = the longest one.																					
Path	Route	T1	T2	T3	T4	T5	T6	T7	T8	T9	T10	T11	T12	T13	T14	T15	Length (days)	Variance	Std Dev	Critical?	
Path A	1→2→6→7→13→14→15	1	1	0	0	0	1	1	0	0	0	0	0	1	1	1	42	10.46	3.234		
Path B	1→2→9→10→14→15	1	1	0	0	0	0	0	0	1	1	0	0	0	1	1	37	8.01	2.830		
Path C	1→3→8→9→10→14→15	1	0	1	0	0	0	0	1	1	1	0	0	0	1	1	45	11.51	3.393		
Path D	1→4→5→13→14→15	1	0	0	1	1	0	0	0	0	0	0	0	1	1	1	39	8.76	2.960		
Path E	1→4→11→12→13→14→15	1	0	0	1	0	0	0	0	0	0	1	1	1	1	1	46	9.81	3.132	CRITICAL	
Project Duration (Critical Path):		46																			
Critical Path Variance (max-variance critical path):		9.81																			
Critical Path Std Dev:		3.132																			

Project Summary — Duration, Cost, Probability		
All values update automatically when you change Crash Days Applied in the Inputs sheet.		
INPUTS		
Target Project Duration (days)	45	← Change to test probability of finishing by other dates
Baseline Project Duration (days)	46	← Original CP duration (set Crash Days = 0 and read from Network Paths)
Indirect Cost per Day Saved (\$)	\$4,200	
Total Indirect Cost (baseline)	\$90,000	
CURRENT PROJECT STATE		
Current Project Duration (days)	46	
Critical Path Variance	9.81	
Critical Path Std Dev	3.132	
PROBABILITY ANALYSIS		
Z-value = (Target – CP Duration) / σ_{CP}	-0.3193	
P(Project ≤ Target Days)	37.48%	
		↑ If this is below 95% for Scenario 1, you need more crashing.
COST ANALYSIS		
Total Normal Direct Cost	\$192,000	
Total Crash Cost Applied	\$0	
TOTAL DIRECT COST	\$192,000	
Days Saved (vs baseline)	0	
Indirect Cost Saved	\$0	
Current Indirect Cost	\$90,000	
TOTAL PROJECT COST (Direct + Indirect)	\$282,000	

Scenario 1: Achieving 95% Confidence by Day 45

Scenario 1 examines the crashing required to raise the probability of completing the project within 45 days to at least 95%. Achieving a 95% confidence level requires a Z-value of at least 1.645, which in turn requires the expected project duration to be reduced substantially below the 45-day target to create sufficient statistical margin against duration variability.

Crashing was applied iteratively, selecting at each step the activity offering the greatest schedule reduction per dollar on the binding critical path. The analysis revealed that the most efficient initial move was to crash Task 15 (Post-Event Feedback and Reporting), which lies on all five paths and carries the lowest crash cost per day at \$2,000; compressing an activity common to every path reduces all path lengths simultaneously and is therefore highly efficient. As crashing proceeded, the critical path migrated: the binding constraint shifted from Path E alone to a combination of Paths C and E at 44 days, and subsequently to Paths A, C, and E at 41 days. This migration illustrates the phenomenon anticipated in the literature, whereby compressing one path causes previously non-critical paths to become binding, requiring continuous re-identification of the critical path.

A further insight concerns the differential value of activities spanning multiple paths. Task 13 (On-Site Logistics Preparation) appears on Paths A, D, and E, so crashing it reduces several path lengths at once, offering greater leverage than activities confined to a single path. The crashing sequence required to reach the target involved compressing Tasks 15, 12, 9, 11, 10, 13, and 3, reducing the project to 39 days. At this duration, the estimated completion probability is approximately 96.8%, satisfying the 95% requirement. The associated crash cost is approximately \$53,000, bringing the total project cost to approximately \$305,600. While this represents an increase of roughly \$23,600 over the baseline cost, it transforms a project with only a 37.5% chance of on-time delivery into one with high confidence of meeting its deadline.

Task Inputs — ANZ Analytics Innovation Conference														
Modify the YELLOW 'Crash Days Applied' column to test scenarios. All other columns auto-calculate.														
Task ID	Task Name	Predecessors	a	m	b	EI (rounded up)	σ (1 dp)	σ^2 (Variance)	Normal Cost (\$)	Max Crash (days)	Crash \$/day	Crash Days Applied	Crashed Duration	Crash Cost (\$)
1	Initial Project Kick-Off	—	1	2	3	2	0.3	0.09	\$3,000	0	—	0	2	\$0
2	Define Conference Themes & Tracks	1	4	6	10	7	1.0	1.00	\$8,000	2	\$4,000	0	7	\$0
3	Keynote Speaker Identification	1	3	6	12	7	1.5	2.25	\$10,000	2	\$5,000	2	5	\$10,000
4	Venue Booking & Contracting	1	5	8	14	9	1.5	2.25	\$18,000	3	\$6,000	0	9	\$0
5	Hybrid Technical Setup	4	6	10	16	11	1.7	2.89	\$22,000	3	\$7,000	0	11	\$0
6	Sponsorship Package Development	2	3	5	9	6	1.0	1.00	\$6,000	1	\$3,000	0	6	\$0
7	Sponsor Outreach & Negotiation	6	5	9	18	10	2.2	4.84	\$12,000	3	\$4,000	0	10	\$0
8	Keynote Speaker Confirmation	3	4	7	13	8	1.5	2.25	\$14,000	2	\$6,000	0	8	\$0
9	Marketing Plan Development	2, 8	4	7	11	8	1.2	1.44	\$9,000	1	\$4,000	1	7	\$4,000
10	Promotional Campaign Execution	9	7	12	20	13	2.2	4.84	\$15,000	3	\$5,000	2	11	\$10,000
11	Catering & Layout Planning	4	6	9	14	10	1.3	1.69	\$11,000	2	\$4,000	2	8	\$8,000
12	Volunteer Recruitment & Training	11	4	7	13	8	1.5	2.25	\$8,000	2	\$3,500	2	6	\$7,000
13	On-Site Logistics Preparation	5, 7, 12	5	9	15	10	1.7	2.89	\$20,000	3	\$6,000	2	8	\$12,000
14	Conference Delivery Day	13, 10	1	1	1	1	0.0	0.00	\$30,000	0	—	0	1	\$0
15	Post-Event Feedback & Reporting	14	3	5	8	6	0.8	0.64	\$6,000	1	\$2,000	1	5	\$2,000
TOTALS									Direct cost → \$192,000				Crash cost → \$53,000	
TOTAL DIRECT COST →									\$245,000					

Project Summary — Duration, Cost, Probability

All values update automatically when you change Crash Days Applied in the Inputs sheet.

INPUTS		
Target Project Duration (days)	45	← Change to test probability of finishing by other dates
Baseline Project Duration (days)	46	← Original CP duration (set Crash Days = 0 and read from Network Paths)
Indirect Cost per Day Saved (\$)	\$4,200	
Total Indirect Cost (baseline)	\$90,000	
CURRENT PROJECT STATE		
Current Project Duration (days)	39	
Critical Path Variance	10.46	
Critical Path Std Dev	3.234	
PROBABILITY ANALYSIS		
Z-value = (Target – CP Duration) / σ_{CP}	1.8552	
P(Project ≤ Target Days)	96.82%	↑ If this is below 95% for Scenario 1, you need more crashing.
COST ANALYSIS		
Total Normal Direct Cost	\$192,000	
Total Crash Cost Applied	\$53,000	
TOTAL DIRECT COST	\$245,000	
Days Saved (vs baseline)	7	
Indirect Cost Saved	\$29,400	
Current Indirect Cost	\$60,600	
TOTAL PROJECT COST (Direct + Indirect)	\$305,600	

Step	Task Crashed	Days Crashed (this step)	Crash \$/day	Step Cost	Cumulative Crash Cost	New CP Duration	P(≤ target)
1	T15	1	\$ 2,000.00	\$ 2,000.00	\$ 2,000.00	45	50%
2	T12	1	\$ 3,500.00	\$ 3,500.00	\$ 5,500.00	44	61.59%
3	T12 + T9	2	3500+4000	\$ 7,500.00	\$ 13,000.00	43	72.22%
4	T11 + T10	2	4000+5000	\$ 9,000.00	\$ 22,000.00	42	81.17%
5	T11 + T10	2	4000+5000	\$ 9,000.00	\$ 31,000.00	41	89.19%
6	T13 + T3	2	6000+5000	\$ 11,000.00	\$ 42,000.00	40	93.89%
7	T13 + T3	2	6000+5000	\$ 11,000.00	\$ 53,000.00	39	96.82%
8							
9							
10							
11							
12							
13							
14							
15							

Scenario 2: Optimum Cost-Time Trade-off

Scenario 2 identifies the schedule that minimises total project cost, balancing the direct cost of crashing against the indirect cost saved per day. The governing decision rule is to continue crashing only while the marginal cost of compressing the schedule by one day remains below the \$4,200 saved in indirect costs for that day.

Applying this rule, only limited crashing proves economically justified. Crashing Task 15 by one day costs \$2,000, and crashing Task 12 by one day costs \$3,500; both fall below the \$4,200 threshold and are therefore worthwhile, reducing the project to 44 days. Beyond this point, however, further compression requires crashing activities on multiple binding paths simultaneously — for example, reducing both Path E and the now-binding Path C — at a combined marginal cost exceeding the \$4,200 indirect saving. At that stage, additional crashing increases total cost rather than reducing it, so the process halts. The cost-time optimum therefore occurs at 44 days, with a total project cost of approximately \$279,100 — marginally below the baseline cost.

It is important to note, however, that this cost-minimising schedule offers only a modest confidence level of on-time completion, estimated at approximately 62%. The cost-time optimum thus minimises expenditure but leaves a substantial residual risk of overrunning the deadline, a tension that is addressed in the strategic recommendations.

Task Inputs — ANZ Analytics Innovation Conference														
Modify the YELLOW 'Crash Days Applied' column to test scenarios. All other columns auto-calculate.														
Task ID	Task Name	Predecessors	a	m	b	ET (rounded up)	σ (1 dp)	σ^2 (Variance)	Normal Cost (\$)	Max Crash (days)	Crash \$/day	Crash Days Applied	Crashed Duration	Crash Cost (\$)
1	Initial Project Kick-Off	—	1	2	3	2	0.3	0.09	\$3,000	0	—	0	2	\$0
2	Define Conference Themes & Tracks	1	4	6	10	7	1.0	1.00	\$8,000	2	\$4,000	0	7	\$0
3	Keynote Speaker Identification	1	3	6	12	7	1.5	2.25	\$10,000	2	\$5,000	0	7	\$0
4	Venue Booking & Contracting	1	5	8	14	9	1.5	2.25	\$18,000	3	\$6,000	0	9	\$0
5	Hybrid Technical Setup	4	6	10	16	11	1.7	2.89	\$22,000	3	\$7,000	0	11	\$0
6	Sponsorship Package Development	2	3	5	9	6	1.0	1.00	\$6,000	1	\$3,000	0	6	\$0
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8	Keynote Speaker Confirmation	3	4	7	13	8	1.5	2.25	\$14,000	2	\$6,000	0	8	\$0
9	Marketing Plan Development	2, 8	4	7	11	8	1.2	1.44	\$9,000	1	\$4,000	0	8	\$0
10	Promotional Campaign Execution	9	7	12	20	13	2.2	4.84	\$15,000	3	\$5,000	0	13	\$0
11	Catering & Layout Planning	4	6	9	14	10	1.3	1.69	\$11,000	2	\$4,000	0	10	\$0
12	Volunteer Recruitment & Training	11	4	7	13	8	1.5	2.25	\$8,000	2	\$3,500	0	8	\$0
13	On-Site Logistics Preparation	5, 7, 12	5	9	15	10	1.7	2.89	\$20,000	3	\$6,000	0	10	\$0
14	Conference Delivery Day	13, 10	1	1	1	1	0.0	0.00	\$30,000	0	—	0	1	\$0
15	Post-Event Feedback & Reporting	14	3	5	8	6	0.8	0.64	\$6,000	1	\$2,000	0	6	\$0
TOTALS									Direct cost → \$192,000				Crash cost →	\$0
									TOTAL DIRECT COST →	\$192,000				

	A	B	C	D	E
1	Project Summary — Duration, Cost, Probability				
2	All values update automatically when you change Crash Days Applied in the Inputs sheet.				
3					
4	INPUTS				
5	Target Project Duration (days)	45	← Change to test probability of finishing by other dates		
6	Baseline Project Duration (days)	46	← Original CP duration (set Crash Days = 0 and read from Network Paths)		
7	Indirect Cost per Day Saved (\$)	\$4,200			
8	Total Indirect Cost (baseline)	\$90,000			
9					
10	CURRENT PROJECT STATE				
11	Current Project Duration (days)	46			
12	Critical Path Variance	9.81			
13	Critical Path Std Dev	3.132			
14					
15	PROBABILITY ANALYSIS				
16	Z-value = (Target – CP Duration) / σ_{CP}	-0.3193			
17	P(Project ≤ Target Days)	37.48%			
18			↑ If this is below 95% for Scenario 1, you need more crashing.		
19					
20	COST ANALYSIS				
21	Total Normal Direct Cost	\$192,000			
22	Total Crash Cost Applied	\$0			
23	TOTAL DIRECT COST	\$192,000			
24	Days Saved (vs baseline)	0			
25	Indirect Cost Saved	\$0			
26	Current Indirect Cost	\$90,000			
27	TOTAL PROJECT COST (Direct + Indirect)	\$282,000			
28					
29					
30					

[illegible]

A "brute-force" approach, crashing every eligible activity to its maximum, achieves the 36-day minimum at a crash cost of approximately \$138,000, giving a total project cost of approximately \$378,000. However, this approach is inefficient, because it expends resources crashing activities that do not affect the binding path. A more economical approach achieves the same 36-day minimum by crashing only those activities necessary to bring all paths down to 36 days, leaving activities on shorter paths uncrashed. This optimised approach reduces the crash cost to approximately \$95,000 and the total cost to approximately \$335,000 — a saving of roughly \$43,000 relative to the brute-force method for an identical schedule outcome.

This scenario yields an important practical insight: when pursuing minimum duration, crashing should be confined to activities that constrain the binding path, since compressing activities on non-binding paths adds cost without shortening the project. It also demonstrates that the minimum-duration schedule is considerably more expensive than the 95% confidence schedule of Scenario 1, raising the question of whether the additional days saved justify their disproportionate marginal cost.

Task Inputs — ANZ Analytics Innovation Conference														
Modify the YELLOW 'Crash Days Applied' column to test scenarios. All other columns auto-calculate.														
Task ID	Task Name	Predecessors	a	m	b	ET (rounded up)	σ (1 dp)	σ^2 (Variance)	Normal Cost (\$)	Max Crash (days)	Crash \$/day	Crash Days Applied	Crashed Duration	Crash Cost (\$)
1	Initial Project Kick-Off	—	1	2	3	2	0.3	0.09	\$3,000	0	—	0	2	\$0
2	Define Conference Themes & Tracks	1	4	6	10	7	1.0	1.00	\$8,000	2	\$4,000	2	5	\$8,000
3	Keynote Speaker Identification	1	3	6	12	7	1.5	2.25	\$10,000	2	\$5,000	2	5	\$10,000
4	Venue Booking & Contracting	1	5	8	14	9	1.5	2.25	\$18,000	3	\$6,000	3	6	\$18,000
5	Hybrid Technical Setup	4	6	10	16	11	1.7	2.89	\$22,000	3	\$7,000	3	8	\$21,000
6	Sponsorship Package Development	2	3	5	9	6	1.0	1.00	\$6,000	1	\$3,000	1	5	\$3,000
7	Sponsor Outreach & Negotiation	6	5	9	18	10	2.2	4.84	\$12,000	3	\$4,000	3	7	\$12,000
8	Keynote Speaker Confirmation	3	4	7	13	8	1.5	2.25	\$14,000	2	\$6,000	2	6	\$12,000
9	Marketing Plan Development	2, 8	4	7	11	8	1.2	1.44	\$9,000	1	\$4,000	1	7	\$4,000
10	Promotional Campaign Execution	9	7	12	20	13	2.2	4.84	\$15,000	3	\$5,000	3	10	\$15,000
11	Catering & Layout Planning	4	6	9	14	10	1.3	1.69	\$11,000	2	\$4,000	2	8	\$8,000
12	Volunteer Recruitment & Training	11	4	7	13	8	1.5	2.25	\$8,000	2	\$3,500	2	6	\$7,000
13	On-Site Logistics Preparation	5, 7, 12	5	9	15	10	1.7	2.89	\$20,000	3	\$6,000	3	7	\$18,000
14	Conference Delivery Day	13, 10	1	1	1	1	0.0	0.00	\$30,000	0	—	0	1	\$0
15	Post-Event Feedback & Reporting	14	3	5	8	6	0.8	0.64	\$6,000	1	\$2,000	1	5	\$2,000
TOTALS									Direct cost →	\$192,000			Crash cost →	\$138,000
TOTAL DIRECT COST →									\$330,000					

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V			
1	Network Paths & Critical Path Identification																								
2	All 5 paths from start (Task 1) to finish (Task 15). Path length = sum of crashed durations along that path. Critical path = the longest one.																								
3																									
4	Path	Route	T1	T2	T3	T4	T5	T6	T7	T8	T9	T10	T11	T12	T13	T14	T15	Length (days)	Variance	Std Dev	Critical?				
5	Path A	1→2→6→7→13→14→15	1	1	0	0	0	1	1	0	0	0	0	0	1	1	1	32	10.46	3.234	CRITICAL				
6	Path B	1→2→9→10→14→15	1	1	0	0	0	0	0	0	1	1	0	0	0	1	1	30	8.01	2.830					
7	Path C	1→3→8→9→10→14→15	1	0	1	0	0	0	0	1	1	1	0	0	0	1	1	36	11.51	3.393					
8	Path D	1→4→5→13→14→15	1	0	0	1	1	0	0	0	0	0	0	0	1	1	1	29	8.76	2.960					
9	Path E	1→4→11→12→13→14→15	1	0	0	1	0	0	0	0	0	0	1	1	1	1	1	35	9.81	3.132					
10																									
11	Project Duration (Critical Path):			36																					
12	Critical Path Variance (max-variance critical path):			11.51																					
13	Critical Path Std Dev:			3.393																					
14																									
15																									

Project Summary — Duration, Cost, Probability		
All values update automatically when you change Crash Days Applied in the Inputs sheet.		
INPUTS		
Target Project Duration (days)	45	← Change to test probability of finishing by other dates
Baseline Project Duration (days)	46	← Original CP duration (set Crash Days = 0 and read from Network Paths)
Indirect Cost per Day Saved (\$)	\$4,200	
Total Indirect Cost (baseline)	\$90,000	
CURRENT PROJECT STATE		
Current Project Duration (days)	36	
Critical Path Variance	11.51	
Critical Path Std Dev	3.393	
PROBABILITY ANALYSIS		
Z-value = (Target – CP Duration) / σ_{CP}	2.6528	
P(Project ≤ Target Days)	99.60%	↑ If this is below 95% for Scenario 1, you need more crashing.
COST ANALYSIS		
Total Normal Direct Cost	\$192,000	
Total Crash Cost Applied	\$138,000	
TOTAL DIRECT COST	\$330,000	
Days Saved (vs baseline)	10	
Indirect Cost Saved	\$42,000	
Current Indirect Cost	\$48,000	
TOTAL PROJECT COST (Direct + Indirect)	\$378,000	

Scenario 4: Accelerated Venue Booking

Scenario 4 conducts a sensitivity analysis on a potential planning change in which venue booking and contracting (Task 4) is expedited, reducing its expected duration from 9 days to 2 days and effectively eliminating its variance. As Task 4 lies on Paths D and E, this change materially alters the network structure. Path E, the original critical path, falls to 39 days, and Path D falls to 32 days. Consequently, the critical path shifts to Path C at 45 days.

A significant outcome of this change is that the probability of on-time completion rises to approximately 50% without any crashing expenditure, since the new critical-path duration of 45 days equals the target. This represents a meaningful improvement over the baseline probability of 37.5%, achieved purely through a structural change rather than additional cost. Moreover, the cost of subsequently reaching higher confidence levels is reduced. The analysis indicates that a 95% confidence level can be achieved for a crash cost of approximately \$33,000 under this scenario, compared with approximately \$53,000 under Scenario 1 — a saving of roughly \$20,000. The revised cost-time optimum under this scenario occurs at 43 days, with a total cost of approximately \$275,400. This scenario therefore demonstrates that a structural change to a single high-duration activity on the critical path can deliver substantial benefits in both schedule confidence and cost, often more efficiently than crashing alone.

Task ID	Task Name	Predecessors	a	m	b	ET (rounded)	σ (1 dp)	σ^2 (Variance)	Normal Cost (\$)	Max Crash (days)	Crash \$/day	Crash Days Applied	Crashed Duration	Crash Cost (\$)
1	Initial Project Kick-Off	—	1	2	3	2	0.3	0.09	\$3,000	0	—	0	2	\$0
2	Define Conference Themes & Tracks	1	4	6	10	7	1.0	1.00	\$8,000	2	\$4,000	0	7	\$0
3	Keynote Speaker Identification	1	3	6	12	7	1.5	2.25	\$10,000	2	\$5,000	0	7	\$0
4	Venue Booking & Contracting	1	2	2	2	0	0.0	0.00	\$18,000	3	\$6,000	0	0	\$0
5	Hybrid Technical Setup	4	6	10	16	11	1.7	2.89	\$22,000	3	\$7,000	0	11	\$0
6	Sponsorship Package Development	2	3	5	9	6	1.0	1.00	\$6,000	1	\$3,000	0	6	\$0
7	Sponsor Outreach & Negotiation	6	5	9	18	10	2.2	4.84	\$12,000	3	\$4,000	0	10	\$0
8	Keynote Speaker Confirmation	3	4	7	13	8	1.5	2.25	\$14,000	2	\$6,000	0	8	\$0
9	Marketing Plan Development	2, 8	4	7	11	8	1.2	1.44	\$9,000	1	\$4,000	0	8	\$0
10	Promotional Campaign Execution	9	7	12	20	13	2.2	4.84	\$15,000	3	\$5,000	0	13	\$0
11	Catering & Layout Planning	4	6	9	14	10	1.3	1.69	\$11,000	2	\$4,000	0	10	\$0
12	Volunteer Recruitment & Training	11	4	7	13	8	1.5	2.25	\$8,000	2	\$3,500	0	8	\$0
13	On-Site Logistics Preparation	5, 7, 12	5	9	15	10	1.7	2.89	\$20,000	3	\$6,000	0	10	\$0
14	Conference Delivery Day	13, 10	1	1	1	1	0.0	0.00	\$30,000	0	—	0	1	\$0
15	Post-Event Feedback & Reporting	14	3	5	8	6	0.8	0.64	\$6,000	1	\$2,000	0	6	\$0
TOTALS									Direct cost → \$192,000				Crash cost →	\$0
									TOTAL DIRECT COST → \$192,000					

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V
Network Paths & Critical Path Identification																					
All 5 paths from start (Task 1) to finish (Task 15). Path length = sum of crashed durations along that path. Critical path = the longest one.																					
Path	Route	T1	T2	T3	T4	T5	T6	T7	T8	T9	T10	T11	T12	T13	T14	T15	Length (days)	Variance	Std Dev	Critical?	
Path A	1→2→6→7→13→14→15	1	1	0	0	0	1	1	0	0	0	0	0	1	1	1	42	10.46	3.234		
Path B	1→2→9→10→14→15	1	1	0	0	0	0	0	0	1	1	0	0	0	1	1	37	8.01	2.830		
Path C	1→3→8→9→10→14→15	1	0	1	0	0	0	0	1	1	1	0	0	0	1	1	45	11.51	3.393	CRITICAL	
Path D	1→4→5→13→14→15	1	0	0	1	1	0	0	0	0	0	0	0	1	1	1	30	6.51	2.551		
Path E	1→4→11→12→13→14→15	1	0	0	1	0	0	0	0	0	0	1	1	1	1	1	37	7.56	2.750		
Project Duration (Critical Path):		45																			
Critical Path Variance (max-variance critical path):		11.51																			
Critical Path Std Dev:		3.393																			

Project Summary — Duration, Cost, Probability		
All values update automatically when you change Crash Days Applied in the Inputs sheet.		
INPUTS		
Target Project Duration (days)	45	← Change to test probability of finishing by other dates
Baseline Project Duration (days)	46	← Original CP duration (set Crash Days = 0 and read from Network Paths)
Indirect Cost per Day Saved (\$)	\$4,200	
Total Indirect Cost (baseline)	\$90,000	
CURRENT PROJECT STATE		
Current Project Duration (days)	45	
Critical Path Variance	11.51	
Critical Path Std Dev	3.393	
PROBABILITY ANALYSIS		
Z-value = (Target – CP Duration) / σ_CP	0.0000	
P(Project ≤ Target Days)	50.00%	↑ If this is below 95% for Scenario 1, you need more crashing.
COST ANALYSIS		
Total Normal Direct Cost	\$192,000	
Total Crash Cost Applied	\$0	
TOTAL DIRECT COST	\$192,000	
Days Saved (vs baseline)	1	
Indirect Cost Saved	\$4,200	
Current Indirect Cost	\$85,800	
TOTAL PROJECT COST (Direct + Indirect)	\$277,800	

Scenario 5: Additional Crashing Capacity for Keynote Confirmation Scenario 5 is a second Scenario 5 examines the effect of increasing the maximum crashing capacity of Task 8 (Keynote Speaker Confirmation) from 2 days to 5 days, reflecting a hypothetical improvement in keynote speakers' response times. Task 8 lies on Path C, the near-critical path.

The analysis indicates that this change has no effect on the cost-time optimum of Scenario 2, on the crashing required to reach the 95% confidence target of Scenario 1, or on the identity of the critical path at those optima. This is because, at those schedules, the additional crashing capacity of Task 8 is not required — the binding constraints lie elsewhere. The only outcome affected is the minimum achievable duration of Scenario 3: with Task 8's expanded capacity, Path C can be compressed further, reducing the overall minimum project duration from 36 days to 35 days, at which point Path E becomes the new binding path. Achieving this 35-day minimum requires a crash cost of approximately \$111,000, giving a total cost of approximately \$346,800.

The practical implication is that the additional flexibility in Task 8 offers value only at the extreme of schedule compression, and even then yields only a single additional day. For the recommended operating range of the project, this flexibility is best retained as a contingency reserve against unforeseen delays rather than deployed pre-emptively, a conclusion reflected in the strategic recommendations.

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Network Paths & Critical Path Identification																				
All 5 paths from start (Task 1) to finish (Task 15). Path length = sum of crashed durations along that path. Critical path = the longest one.																				
Path	Route	T1	T2	T3	T4	T5	T6	T7	T8	T9	T10	T11	T12	T13	T14	T15	Length (days)	Variance	Std Dev	Critical?
Path A	1→2→6→7→13→14→15	1	1	0	0	0	1	1	0	0	0	0	0	1	1	1	42	10.46	3.234	
Path B	1→2→9→10→14→15	1	1	0	0	0	0	0	0	1	1	0	0	0	1	1	37	8.01	2.830	
Path C	1→3→8→9→10→14→15	1	0	1	0	0	0	0	1	1	1	0	0	0	1	1	45	11.51	3.393	
Path D	1→4→5→13→14→15	1	0	0	1	1	0	0	0	0	0	0	0	1	1	1	39	8.76	2.960	
Path E	1→4→11→12→13→14→15	1	0	0	1	0	0	0	0	0	0	1	1	1	1	1	46	9.81	3.132	CRITICAL
Project Duration (Critical Path):		46																		
Critical Path Variance (max-variance critical path):		9.81																		
Critical Path Std Dev:		3.132																		

Project Summary — Duration, Cost, Probability		
All values update automatically when you change Crash Days Applied in the Inputs sheet.		
INPUTS		
Target Project Duration (days)	45	← Change to test probability of finishing by other dates
Baseline Project Duration (days)	46	← Original CP duration (set Crash Days = 0 and read from Network Paths)
Indirect Cost per Day Saved (\$)	\$4,200	
Total Indirect Cost (baseline)	\$90,000	
CURRENT PROJECT STATE		
Current Project Duration (days)	46	
Critical Path Variance	9.81	
Critical Path Std Dev	3.132	
PROBABILITY ANALYSIS		
Z-value = (Target – CP Duration) / σ_{CP}	-0.3193	
P(Project ≤ Target Days)	37.48%	↑ If this is below 95% for Scenario 1, you need more crashing.
COST ANALYSIS		
Total Normal Direct Cost	\$192,000	
Total Crash Cost Applied	\$0	
TOTAL DIRECT COST	\$192,000	
Days Saved (vs baseline)	0	
Indirect Cost Saved	\$0	
Current Indirect Cost	\$90,000	
TOTAL PROJECT COST (Direct + Indirect)	\$282,000	

7. Strategic Recommendations

7.1 Recommended Strategy

Drawing on the five scenarios analysed above, this report recommends that the Business School adopt a hybrid strategy that prioritises a low-cost structural change to the project plan, targets a high-confidence completion schedule rather than the lowest-cost or shortest-possible schedule, and holds a defined contingency in reserve. Under the baseline plan, the project is expected to take 46 days, with only a 37.5% probability of meeting the 45-day deadline — a level of confidence that the report considers insufficient for an initiative of this strategic importance.

7.2 Primary Recommendation: Pursue Expedited Venue Booking (Task 4)

The single most valuable action available to the Business School is to secure expedited approval for venue booking and contracting, as modelled in Scenario 4. Reducing the expected duration of Task 4 shifts the critical path and raises the probability of on-time completion to approximately 50% without any crashing expenditure. More importantly, it lowers the cost of subsequently reaching a 95% confidence level: the analysis indicates that the 95% target can then be achieved for a crash cost of around \$33,000, compared with approximately \$53,000 under the original plan (Scenario 1) — a saving of roughly \$20,000. Because this change depends primarily on administrative priority rather than additional spending, it represents the project's highest-leverage opportunity and should be pursued first.

7.3 Recommended Target: 95% Confidence, Not Cost-Minimum

The report recommends targeting the 95% confidence schedule rather than the cost-time optimum identified in Scenario 2. While the cost-time optimum (44 days, total cost approximately \$279,100) is marginally cheaper, it carries only an estimated 62% probability of meeting the deadline. For a flagship conference with fixed external commitments, this residual risk of late delivery is not justified by the modest saving. Equally, the report advises against pursuing the minimum-duration schedule of Scenario 3, where the marginal cost of each additional day saved becomes disproportionately high and is not warranted once a robust confidence level has been achieved.

7.4 Risk Mitigation and Contingency

Three schedule vulnerabilities warrant active monitoring. First, Task 10 is both the longest activity and one of the most variable, making it the largest single source of completion-time uncertainty. Second, Path C operates with minimal slack at baseline and becomes critical as the schedule is compressed, so it should be tracked as a near-critical path throughout execution. Third, the additional crashing flexibility for keynote speaker confirmation (Task 8), examined in Scenario 5, should be retained as a contingency reserve rather than deployed pre-emptively, providing a buffer against unforeseen delays consistent with the buffer-management principle discussed earlier (Goldratt, 1997).

7.5 Summary

In combination, these recommendations enable the Business School to deliver the conference with high confidence at a cost materially below that of the original 95% plan, while preserving flexibility to respond to emerging risks.

Section 8: Appendices

Appendix A : Activity Data and Three-Point Estimates

Task Inputs — ANZ Analytics Innovation Conference															
Modify the YELLOW 'Crash Days Applied' column to test scenarios. All other columns auto-calculate.															
Task ID	Task Name	Predecessors	a	m	b	ET (rounded up)	σ (1 dp)	σ^2 (Variance)	Normal Cost (\$)	Max Crash (days)	Crash \$/day	Crash Days Applied	Crashed Duration	Crash Cost (\$)	
1	Initial Project Kick-Off	—	1	2	3	2	0.3	0.09	\$3,000	0	—	0	2	\$0	
2	Define Conference Themes & Tracks	1	4	6	10	7	1.0	1.00	\$8,000	2	\$4,000	0	7	\$0	
3	Keynote Speaker Identification	1	3	6	12	7	1.5	2.25	\$10,000	2	\$5,000	0	7	\$0	
4	Venue Booking & Contracting	1	5	8	14	9	1.5	2.25	\$18,000	3	\$6,000	0	9	\$0	
5	Hybrid Technical Setup	4	6	10	16	11	1.7	2.89	\$22,000	3	\$7,000	0	11	\$0	
6	Sponsorship Package Development	2	3	5	9	6	1.0	1.00	\$6,000	1	\$3,000	0	6	\$0	
7	Sponsor Outreach & Negotiation	6	5	9	18	10	2.2	4.84	\$12,000	3	\$4,000	0	10	\$0	
8	Keynote Speaker Confirmation	3	4	7	13	8	1.5	2.25	\$14,000	2	\$6,000	0	8	\$0	
9	Marketing Plan Development	2, 8	4	7	11	8	1.2	1.44	\$9,000	1	\$4,000	0	8	\$0	
10	Promotional Campaign Execution	9	7	12	20	13	2.2	4.84	\$15,000	3	\$5,000	0	13	\$0	
11	Catering & Layout Planning	4	6	9	14	10	1.3	1.69	\$11,000	2	\$4,000	0	10	\$0	
12	Volunteer Recruitment & Training	11	4	7	13	8	1.5	2.25	\$8,000	2	\$3,500	0	8	\$0	
13	On-Site Logistics Preparation	5, 7, 12	5	9	15	10	1.7	2.89	\$20,000	3	\$6,000	0	10	\$0	
14	Conference Delivery Day	13, 10	1	1	1	1	0.0	0.00	\$30,000	0	—	0	1	\$0	
15	Post-Event Feedback & Reporting	14	3	5	8	6	0.8	0.64	\$6,000	1	\$2,000	0	6	\$0	
TOTALS									Direct cost → \$192,000				Crash cost →	\$0	
TOTAL DIRECT COST →									\$192,000						

Appendix B : PERT Calculation Method

Expected Time (ET):

$$ET = (a + 4m + b) / 6, \text{ rounded up to the nearest whole day}$$

where a = optimistic time

m = most likely time

b = pessimistic time

Activity Variance (σ^2): $\sigma^2 = [(b - a) / 6]^2$

Standard Deviation (σ): $\sigma = (b - a) / 6$

Critical Path Variance:

$\sigma^2(\text{CP})$ = sum of variances of all activities on the critical path

Probability of completion within target time T:

$Z = (T - \mu) / \sigma(\text{CP})$

$P = \text{NORMDIST}(Z)$

where μ = expected critical path duration

$\sigma(\text{CP})$ = standard deviation of the critical path

Example : This is the example of Task1:ET, Standard Deviation,Activity Variance

Task Inputs — ANZ Analytics Innovation Conference															
Modify the YELLOW 'Crash Days Applied' column to test scenarios. All other columns auto-calculate.															
Task ID	Task Name	Predecessors	a	m	b	ET (rounded)	σ (1 dp)	σ^2 (Variance)	Normal Cost (\$)	Max Crash (days)	Crash \$/day	Crash Days Applied	Crashed Duration	Crash Cost (\$)	
1	Initial Project Kick-Off	—	1	2	3	2	0.3	0.09	\$3,000	0	—	0	2	\$0	
2	Define Conference Themes & Tracks	1	4	6	10	7	1.0	1.00	\$8,000	2	\$4,000	0	7	\$0	
3	Keynote Speaker Identification	1	3	6	12	7	1.5	2.25	\$10,000	2	\$5,000	0	7	\$0	
4	Venue Booking & Contracting	1	5	8	14	9	1.5	2.25	\$18,000	3	\$6,000	0	9	\$0	
5	Hybrid Technical Setup	4	6	10	16	11	1.7	2.89	\$22,000	3	\$7,000	0	11	\$0	
6	Sponsorship Package Development	2	3	5	9	6	1.0	1.00	\$6,000	1	\$3,000	0	6	\$0	
7	Sponsor Outreach & Negotiation	6	5	9	18	10	2.2	4.84	\$12,000	3	\$4,000	0	10	\$0	
8	Keynote Speaker Confirmation	3	4	7	13	8	1.5	2.25	\$14,000	2	\$6,000	0	8	\$0	
9	Marketing Plan Development	2, 8	4	7	11	8	1.2	1.44	\$9,000	1	\$4,000	0	8	\$0	
10	Promotional Campaign Execution	9	7	12	20	13	2.2	4.84	\$15,000	3	\$5,000	0	13	\$0	
11	Catering & Layout Planning	4	6	9	14	10	1.3	1.69	\$11,000	2	\$4,000	0	10	\$0	
12	Volunteer Recruitment & Training	11	4	7	13	8	1.5	2.25	\$8,000	2	\$3,500	0	8	\$0	
13	On-Site Logistics Preparation	5, 7, 12	5	9	15	10	1.7	2.89	\$20,000	3	\$6,000	0	10	\$0	
14	Conference Delivery Day	13, 10	1	1	1	1	0.0	0.00	\$30,000	0	—	0	1	\$0	
15	Post-Event Feedback & Reporting	14	3	5	8	6	0.8	0.64	\$6,000	1	\$2,000	0	6	\$0	
TOTALS															
Direct cost →									\$192,000				Crash cost →	\$0	
TOTAL DIRECT COST →									\$192,000						

Task Inputs — ANZ Analytics Innovation Conference															
Modify the YELLOW 'Crash Days Applied' column to test scenarios. All other columns auto-calculate.															
Task ID	Task Name	Predecessors	a	m	b	ET (rounded)	σ (1 dp)	σ^2 (Variance)	Normal Cost (\$)	Max Crash (days)	Crash \$/day	Crash Days Applied	Crashed Duration	Crash Cost (\$)	
1	Initial Project Kick-Off	—	1	2	3	2	0.3	0.09	\$3,000	0	—	0	2	\$0	
2	Define Conference Themes & Tracks	1	4	6	10	7	1.0	1.00	\$8,000	2	\$4,000	0	7	\$0	
3	Keynote Speaker Identification	1	3	6	12	7	1.5	2.25	\$10,000	2	\$5,000	0	7	\$0	
4	Venue Booking & Contracting	1	5	8	14	9	1.5	2.25	\$18,000	3	\$6,000	0	9	\$0	
5	Hybrid Technical Setup	4	6	10	16	11	1.7	2.89	\$22,000	3	\$7,000	0	11	\$0	
6	Sponsorship Package Development	2	3	5	9	6	1.0	1.00	\$6,000	1	\$3,000	0	6	\$0	
7	Sponsor Outreach & Negotiation	6	5	9	18	10	2.2	4.84	\$12,000	3	\$4,000	0	10	\$0	
8	Keynote Speaker Confirmation	3	4	7	13	8	1.5	2.25	\$14,000	2	\$6,000	0	8	\$0	
9	Marketing Plan Development	2, 8	4	7	11	8	1.2	1.44	\$9,000	1	\$4,000	0	8	\$0	
10	Promotional Campaign Execution	9	7	12	20	13	2.2	4.84	\$15,000	3	\$5,000	0	13	\$0	
11	Catering & Layout Planning	4	6	9	14	10	1.3	1.69	\$11,000	2	\$4,000	0	10	\$0	
12	Volunteer Recruitment & Training	11	4	7	13	8	1.5	2.25	\$8,000	2	\$3,500	0	8	\$0	
13	On-Site Logistics Preparation	5, 7, 12	5	9	15	10	1.7	2.89	\$20,000	3	\$6,000	0	10	\$0	
14	Conference Delivery Day	13, 10	1	1	1	1	0.0	0.00	\$30,000	0	—	0	1	\$0	
15	Post-Event Feedback & Reporting	14	3	5	8	6	0.8	0.64	\$6,000	1	\$2,000	0	6	\$0	
TOTALS															
Direct cost →									\$192,000				Crash cost →	\$0	
TOTAL DIRECT COST →									\$192,000						

SUM														
Task Inputs — ANZ Analytics Innovation Conference														
Modify the YELLOW 'Crash Days Applied' column to test scenarios. All other columns auto-calculate.														
Task ID	Task Name	Predecessors	a	m	b	ET (rounded)	σ (1 dp)	σ^2 (Variance)	Normal Cost (\$)	Max Crash (days)	Crash \$/day	Crash Days Applied	Crashed Duration	Crash Cost (\$)
1	Initial Project Kick-Off	—	1	2	3	2	0.3	$=H5^2$	\$3,000	0	—	0	2	\$0
2	Define Conference Themes & Tracks	1	4	6	10	7	1.0	1.00	\$8,000	2	\$4,000	0	7	\$0
3	Keynote Speaker Identification	1	3	6	12	7	1.5	2.25	\$10,000	2	\$5,000	0	7	\$0
4	Venue Booking & Contracting	1	5	8	14	9	1.5	2.25	\$18,000	3	\$6,000	0	9	\$0
5	Hybrid Technical Setup	4	6	10	16	11	1.7	2.89	\$22,000	3	\$7,000	0	11	\$0
6	Sponsorship Package Development	2	3	5	9	6	1.0	1.00	\$6,000	1	\$3,000	0	6	\$0
7	Sponsor Outreach & Negotiation	6	5	9	18	10	2.2	4.84	\$12,000	3	\$4,000	0	10	\$0
8	Keynote Speaker Confirmation	3	4	7	13	8	1.5	2.25	\$14,000	2	\$6,000	0	8	\$0
9	Marketing Plan Development	2, 8	4	7	11	8	1.2	1.44	\$9,000	1	\$4,000	0	8	\$0
10	Promotional Campaign Execution	9	7	12	20	13	2.2	4.84	\$15,000	3	\$5,000	0	13	\$0
11	Catering & Layout Planning	4	6	9	14	10	1.3	1.69	\$11,000	2	\$4,000	0	10	\$0
12	Volunteer Recruitment & Training	11	4	7	13	8	1.5	2.25	\$8,000	2	\$3,500	0	8	\$0
13	On-Site Logistics Preparation	5, 7, 12	5	9	15	10	1.7	2.89	\$20,000	3	\$6,000	0	10	\$0
14	Conference Delivery Day	13, 10	1	1	1	1	0.0	0.00	\$30,000	0	—	0	1	\$0
15	Post-Event Feedback & Reporting	14	3	5	8	6	0.8	0.64	\$6,000	1	\$2,000	0	6	\$0
TOTALS									Direct cost → \$192,000			Crash cost →		\$0
									TOTAL DIRECT COST →					\$192,000

Appendix C: Network Paths and Critical Path Identification

Network Paths & Critical Path Identification																			
All 5 paths from start (Task 1) to finish (Task 15). Path length = sum of crashed durations along that path. Critical path = the longest one.																			
Path	Route	T1	T2	T3	T4	T5	T6	T7	T8	T9	T10	T11	T12	T13	T14	T15	Length (days)	Variance	Std Dev
Path A	1→2→6→7→13→14→15	1	1	0	0	0	1	1	0	0	0	0	0	1	1	1	42	10.46	3.234
Path B	1→2→9→10→14→15	1	1	0	0	0	0	0	0	1	1	0	0	0	1	1	37	8.01	2.830
Path C	1→3→8→9→10→14→15	1	0	1	0	0	0	0	1	1	1	0	0	0	1	1	45	11.51	3.393
Path D	1→4→5→13→14→15	1	0	0	1	1	0	0	0	0	0	0	0	1	1	1	39	8.76	2.960
Path E	1→4→11→12→13→14→15	1	0	0	1	0	0	0	0	0	0	1	1	1	1	1	46	9.81	3.132
Project Duration (Critical Path):				46															
Critical Path Variance (max-variance critical path):				9.81															
Critical Path Std Dev:				3.132															

Appendix D:Baseline Probability Calculation

1 **Project Summary — Duration, Cost, Probability**

2 All values update automatically when you change Crash Days Applied in the Inputs sheet.

3

4

5 **INPUTS**

6 Target Project Duration (days)45← Change to test probability of finishing by other dates

7 Baseline Project Duration (days)46← Original CP duration (set Crash Days = 0 and read from Network Paths)

8 Indirect Cost per Day Saved (\$) \$4,200

9 Total Indirect Cost (baseline) \$90,000

10

11 **CURRENT PROJECT STATE**

12 Current Project Duration (days)46

13 Critical Path Variance9.81

14 Critical Path Std Dev3.132

15

16 **PROBABILITY ANALYSIS**

17 Z-value = (Target – CP Duration) / σ_CP-0.3193

18 P(Project ≤ Target Days)37.48%

19

20

21 **COST ANALYSIS**

22 Total Normal Direct Cost\$192,000

23 Total Crash Cost Applied\$0

24 TOTAL DIRECT COST\$192,000

25 Days Saved (vs baseline)0

26 Indirect Cost Saved\$0

27 Current Indirect Cost\$90,000

28 TOTAL PROJECT COST (Direct + Indirect)\$282,000

29

30

31

32

33

34

35

36

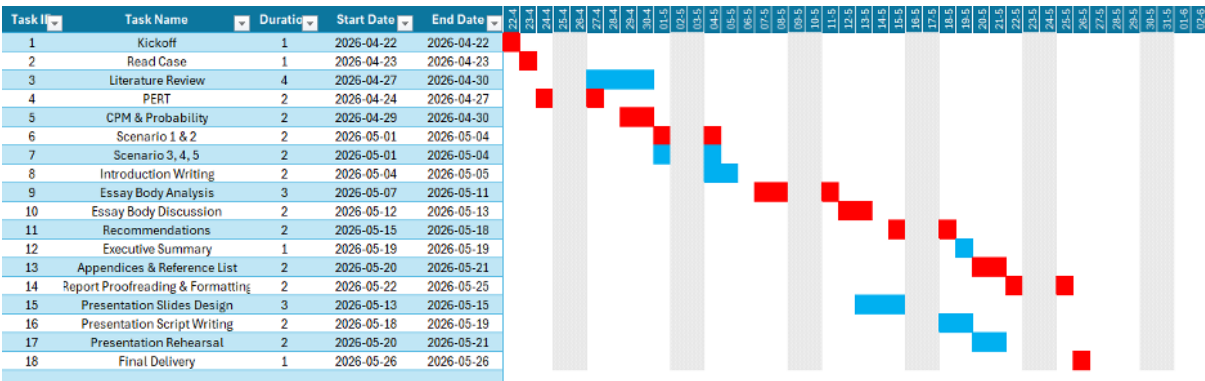
Appendix E:Scenario Crashing Logs

1	Crashing Decision Log							
2	<i>Record your step-by-step crashing decisions here. Use one row per crash step. At each step: identify the cheapest crashable task on the critical path, crash it 1 day.</i>							
3								
4								
5								
6								
7								
8								
9								
10								
11	Step	Task Crashed	Days Crashed (this step)	Crash \$/day	Step Cost	Cumulative Crash Cost	New CP Duration	P(≤ target)
12	1	T15	1	\$ 2,000.00	\$ 2,000.00	\$ 2,000.00	45	50%
13	2	T12	1	\$ 3,500.00	\$ 3,500.00	\$ 5,500.00	44	61.59%
14	3	T12 + T9	2	3500+4000	\$ 7,500.00	\$ 13,000.00	43	72.22%
15	4	T11 + T10	2	4000+5000	\$ 9,000.00	\$ 22,000.00	42	81.17%
16	5	T11 + T10	2	4000+5000	\$ 9,000.00	\$ 31,000.00	41	89.19%
17	6	T13 + T3	2	6000+5000	\$ 11,000.00	\$ 42,000.00	40	93.89%
18	7	T13 + T3	2	6000+5000	\$ 11,000.00	\$ 53,000.00	39	96.82%
19	8							
20	9							
21	10							
22	11							
23	12							
24	13							
25	14							
26	15							
27								
28								
29								

Appendix F:Cost-Time Trade-off Table

	A	B	C	D	E	F
1	Cost–Time Tradeoff Table (for Scenario 2)					
2	Fill this in as you crash. For each project duration achievable, record the direct and indirect costs. Total cost minimum = optimum point.					
3						
4	Project Duration (days)	Cumulative Crash Cost (\$)	Total Direct Cost (\$)	Indirect Cost (\$)	Total Cost (\$)	Notes
5	46	\$0	\$192,000	\$90,000	\$282,000	
6	45	\$2,000	\$194,000	\$85,800	\$279,800	
7	44	\$5,500	\$197,500	\$81,600	\$279,100	
8	43	\$13,000	\$205,000	\$77,400	\$282,400	
9	42	\$22,000	\$214,000	\$73,200	\$287,200	
0	41	\$31,000	\$223,000	\$69,000	\$292,000	
1	40	\$42,000	\$234,000	\$64,800	\$298,800	
2	39	\$53,000	\$245,000	\$60,600	\$305,600	
3						
4						
5						
6						
7						
8						
9				Minimum Total Cost:	\$279,100	
0				Optimum Project Duration (days):	44	
1						

Appendix G:Baseline Gantt Chart



Appendix H:Scenario Results Summary

Scenario	Objective	Duration (days)	P(≤45)	Crash Cost	Total Cost
Baseline	No crashing	46	37.5%	\$0	\$282,000
Scenario 1	95% confidence	39	~96.8%	\$53,000	\$305,600
Scenario 2	Cost-time optimum	44	~62%	\$5,500	\$279,100
Scenario 3	Min duration (optimised)	36	~100%	\$95,000	\$335,000
Scenario 4	Expedited Task 4 + 95%	43	~95%	\$33,000	\$275,400
Scenario 5	Extra Task 8	35	~100%	\$111,000	\$346,800

crash (min
dur.)

Section 9:References

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